

Lecture 02-Part 1

Literature Search + Review

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Today's lecture

- Highlight the key parts of the readings
- “Primary vs Secondary Research”
- Taxonomy of scientific articles
- Conference vs Journal publications in Computer Science
- The different types of “research paper/article”
- How to find “good quality” research papers
- Discuss the written assessed exercise

Reading 1:

Reading + Summarising Papers

What's important in the reading - “reading + summarising”

Very Important:

- Learn **how to structure of a paper**

There is an expected structure to writing a research paper

- Papers can/do differ from this format
- However, the vast majority papers follow this structure
- If it doesn't many readers will say “what's with this paper structure”

There are templates and conventions to follow

What's important in the reading - “reading + summarising”

Very Important:

- **Learn how to structure of a paper**

Abstract -> Introduction -> Background -> Methods/Design -> Results -> Discussion -> Conclusion

- **Discussion includes discussion of results , limitations**
- **Limitations and future work could be in the conclusion**

What's important in the reading - “reading + summarising”

Very Important:

- **Learn how to read a paper**

You don't just “read page 1, read page 2, ...”

- **Read the Title/Abstract -> Introduction/Conclusion -> Skim The Rest**
- **Skimming is essential skill / part of reading papers**
- **NOTE: Some papers you don't have to read beyond a skim**

What's important in the reading - “reading + summarising”

Very Important:

- Learn **how to read a paper**

Not all papers are worth reading fully:

- Some make small / incremental contributions to work you already know
- Determining what is an “important paper” vs an “unimportant paper” is itself a valuable skill to practice
- We will talk later about how to judge the quality of the paper

Although one that is developed over time as you read more and more papers

What's important in the reading - “reading + summarising”

Very Important:

- **Learn how to summarise a paper (and keep a record of this)**

You don't want to be in the position to have to re-read a lot of papers

- **Some you will want to re-read (very important papers, work you are building directly on)**
- **But many you will just want a quick reminder before you reference it**

What's important in the reading - “reading + summarising”

Very Important:

- Learn **how to critique a paper**

This is an essential part of doing research

- There are expectations for how a paper is reviewed

What's important in the reading - “reading + summarising”

In short:

- Learn how to **structure a paper**
- Learn how to **read a paper**
- Learn how to **summarise a paper (and keep a record of this)**
- Learn how to **critique a paper**

Reading 2 : Literature Review

What is the difference between a summary and a literature review ?

What's important in the reading - “literature review”

Very Important:

- Learn **how to reference/cite papers**

You are required to reference/cite prior works when reporting science

- To contextualise and motivate your work
- To justify design decisions that you've made
- To discuss your results relative to past works

There are different citation styles depends on the template

What's important in the reading - “literature review”

Very Important:

- **Learn how to structure/write a literature review**

There is an expected structure to writing a literature review

- But there's some guidelines you can follow to make your life easier

What's important in the reading - “literature review”

Very Important:

- **Learn how to find research papers**

Even something as simple as “finding papers” can be done systematically

- **We'll talk about this today also**

What's important in the reading - “literature review”

In short:

- Learn how to **reference/cite papers**
- Learn how to **structure/write a literature review**
- Learn how to **find research papers**

Today's lecture

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Primary vs Secondary Research

Primary research

Research conducted to gather first-hand information for the current research problem

- Based on raw data – collected by the researcher themselves – but specific to the needs of the researcher
- Process is very involved, high cost, and time intensive

Secondary research

Research using information/data gathered originally by a primary research

- Based on existing data – collected by someone else – so may or may not be specific to the needs of the researcher
- Process is easier, low cost, and less time intensive

Note: This is the analysis and interpretation of data collected by someone else who has conducted primary research

Some data sets are really expensive to capture

Sometimes the contribution of a research project is the data itself

- **A lot of time and money spent obtaining the data**
- **Idea being to release it for others to use**

Some data is extremely difficult to capture

- **Extremes of “very rare resource/access” vs “massive data sets about a particular topic”**

Some data is harder to share .. Ethical considerations!

As researchers we are limited

Researchers are often limited by what we have money/time to do

- Sometimes secondary research is the only way the research can be done

There's an element of research field/topic (research question) as well:

- An HCI researcher: mostly do primary research
- A machine learning cancer researcher: mostly do secondary research
- Sometimes they are used together

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**Not all research papers
are of equal scope**

Some research papers are more equal than others

As a researcher you will work on problems of a variety of scope

- **This variety of research will produce a variety of paper types**

Not every paper will change the world (few will ever do that)

- **Important to know your scope so you can appropriately present / sell the work to other researchers**



Articles

About 3,440,000 results (0.07 sec)

Any time

Since 2022

Since 2021

Since 2018

Custom range...

Sort by relevance

Sort by date

Any type

Review articles

☐ include patents☒ include citations☒ Create alertResearch on presence in **virtual reality**: A survey[M.J. Schuemie](#), [P. Van Der Straaten](#), M. Krijn... - *CyberPsychology & ...*, 2001 - [liebertpub.com](#)

... **Virtual Reality** (VR) is starting to be used in psychological ... and for mental models of the **virtual** space. A review of the ... such as emotional responses to **virtual** stimuli. An investigation ...

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... Understanding **Virtual Reality** arrives at a time when the technologies behind **virtual reality** have advanced to the point that it is possible to develop and deploy meaningful ...

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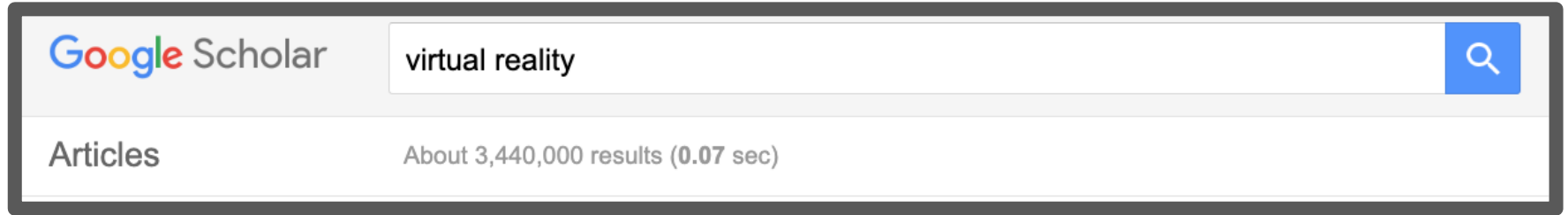
Related searches

[augmented virtual reality](#)[immersive virtual reality](#)[virtual reality presence](#)[virtual reality exposure therapy](#)[virtual reality sickness](#)[virtual reality simulator](#)[virtual reality stroke rehabilitation](#)[haptic virtual reality](#)State of the art of **virtual reality** technology[C Anthes](#), [RJ García-Hernández](#)... - 2016 IEEE ... , 2016 - [ieeexplore.ieee.org](#)

... Current development in **Virtual Reality** (VR) technology is happening at unprecedented ... 't until 1989 when Jaron Lanier coined the term **Virtual Reality** [2] trying to aggregate the different ...

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Searching google scholar for “virtual reality”

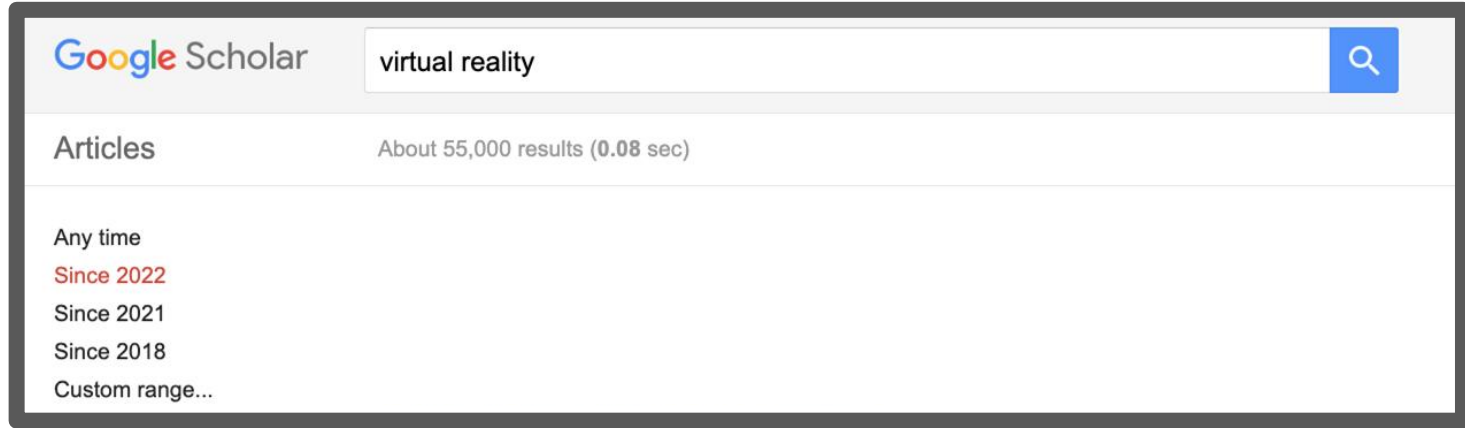


The image shows a screenshot of the Google Scholar search interface. The search bar contains the text "virtual reality". To the left of the search bar is the "Google Scholar" logo. To the right of the search bar is a blue button with a magnifying glass icon. Below the search bar, the word "Articles" is displayed on the left, and "About 3,440,000 results (0.07 sec)" is displayed on the right.

Google Scholar	virtual reality	
Articles	About 3,440,000 results (0.07 sec)	

NOTE: Demonstrative example made two years ago

Searching google scholar for “virtual reality” published this year



A screenshot of the Google Scholar search interface. The search bar at the top contains the text "virtual reality" and a blue search button with a magnifying glass icon. Below the search bar, the text "Articles" is displayed on the left, and "About 55,000 results (0.08 sec)" is displayed on the right. Below this, there is a list of filters: "Any time", "Since 2022" (highlighted in red), "Since 2021", "Since 2018", and "Custom range...".

NOTE: Demonstrative example two years ago

Taxonomy of research papers

A taxonomy of research papers

Breakthrough: solves a long-standing problem, or introduces a totally new way of looking at the world

Ground-breaking: opens up a field that has not been well explored or understood, and lays a firm foundation

Progress: raises and solves important new problems in a well-established field

“A Guide for New Referees in Theoretical Computer Science” by Ian Parberry. 1989

A taxonomy of research paper

Reprise: reworks an existing result (by the authors or others), yielding new insight

Tinkering: extends a known result by a more careful, but non-obvious, methodology

Debugging: elucidates and repairs a flawed previously-published result

“A Guide for New Referees in Theoretical Computer Science” by Ian Parberry. 1989

A taxonomy of research paper

Survey: surveys and unifies a body of knowledge

Applications: solves some problem in a practitioner's context or reports on the development of a new system, usually in a research context

“A Guide for New Referees in Theoretical Computer Science” by Ian Parberry. 1989

A taxonomy of research papers

- Breakthrough
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- Survey
- Applications

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A taxonomy of research papers

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- Survey
- Applications



You will probably spend most of your time working on those!

As a researcher...

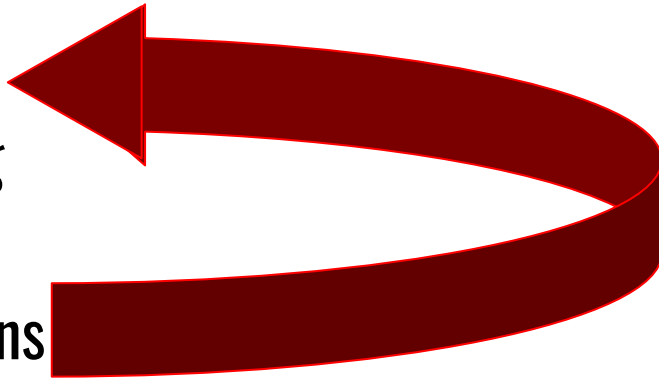
- Breakthrough
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- Applications



If your “Progress” work keeps on getting rejected you’ll eventually claim it was always this

As a researcher...

- Breakthrough
- Ground-breaking
- Progress
- Reprise
- Tinkering
- Debugging
- Survey
- Applications



Same for any repeatedly rejected “Applications” work

As a researcher...

- Breakthrough
- Ground-breaking
- Progress
- Reprise
- Tinkering
- Debugging
- Survey
- Applications



You'll probably write at least 1 or 2 of these

As a researcher...

- Breakthrough
- Ground-breaking
- Progress
- Reprise
- Tinkering
- Debugging
- Survey
- Applications



Hopefully you never need to write this for one of your own papers / work

As a researcher...

- Breakthrough
- Ground-breaking
- Progress
- Reprise
- Tinkering
- Debugging
- Survey
- Applications



Some of these terms are not commonly used nowadays

As a researcher...

- Breakthrough
- Ground-breaking
- Progress
- Reprise
- Tinkering
- Debugging
- Survey
- Applications



This is what you'll probably spend your career in hope of achieving

As a researcher...

- Breakthrough
- Ground-breaking
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- Applications



It will need a big deal of luck and intelligence!

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Conferences vs Journals in Computer Science research

Computer science is a conference dominated field

Unlike other fields (where journal publications are the dominant publication) computer science is conference publication oriented

- Conference papers are much shorter (6-12 pages) than journal papers
- A faster submission-publication process preferred by experimentalists (e.g. 7 months opposed to 24 months)

NOTE: There are journal publications in CS too it's just the majority chase conference papers

Some argue this “conference dominance” should change

Time for Computer Science to Grow Up

Lance Fortnow
Northwestern University

Abstract

Unlike every other academic field, Computer Science uses conferences rather than journals as the main publication venue. While this made sense for a young discipline, our field has matured and the conference model has fractured the discipline and skewed it towards short-term deadline-driven research.

Computer Science should refocus the conference system on its primary purpose of bringing researchers together. We should use archive sites as the main method of quick paper dissemination and the journal system as the vehicle for advancing ones reputation.

NOTE: This article was over 10 years old

It's unlikely to change

How conferences work might but it seems unlikely it'll switch to journal oriented

**So computer science
research is predominantly
conference publications**



A quick note about publishing works

You've conducted the research — what next?

Once you've conducted your research and written it up as a paper you'll look to submit your work to a peer reviewed publication

- In some topics works are also posted on arXiv at this time**
- But often the work is submitted to a conference/journal for review**

You've conducted the research — what next?

How the paper is reviewed depends on the publication:

- Can vary by number of reviewers, review template given to reviewers, if there's a revise and resubmission process, etc

Ultimately though after some time the paper will either be accepted or rejected for publication at the submitted venue

If published at a conference often you are expected to attend to present the work in a talk

You've conducted the research — what next?

NOTE: There are rules surrounding publication

- **You can't submit the same work to multiple publications at once**
 - **This is partly what makes this step slow**
- **Often the work must be fully anonymous and follow the template guidelines of the publication being submit to**

A quick note about publishing works

How to pick where to submit work?

The decision must also be made regarding where work will be published:

- Not all conferences are equal — ranking, scope, community

Other decisions also play into this — maybe you know some organisers and want to attend to try and collaborate with them on a future project, maybe the venue is in an expensive location and you don't have the budget to attend

How to pick where to submit work?

Conferences have a prestige associated with them

- Lower acceptance rates + higher expected quality of work = more prestigious conference venue

As a researcher you want to publish at the most prestigious venues in your field

- Looks good on job/grant applications

Ranking of conference prestige

CORE Ranking (one of several conference ranking services)

CORE2021 Summary:

A* - 7.24% of 801 ranked venues

A - 16.1% of 801 ranked venues

B - 37.08% of 801 ranked venues

Australasian B - 1.62% of 801 ranked venues

C - 36.2% of 801 ranked venues

Australasian C - 1.75% of 801 ranked venues

Other - 167 total

There are options to “pay to win”

Predatory journals do exist

- **Very low standards for entry**
- **Very high fees for publication**

A “red flag” to have such publications on your record for some employers and funding bodies

There are options to “pay to win”

Predatory journals are likely to possess the following characteristics:

1. Wide journal scope
2. Not indexed in databases
3. Make false claims regarding impact factor
4. Claim low author-submission fees
5. Extremely high acceptance rate
6. Accelerated publication process
7. Very short peer-review period
8. Inconsistencies in publication periods
9. Small or unknown editorial board
10. Editorial board members are not experts in the journal topics
11. Unprofessional, unsecured website
12. Errors in titles and journal descriptions
13. Asks authors to submit articles via email or a simple form

There are options to “pay to win”

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Will publish basically anything that's somewhat coherent

There are options to “pay to win”

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Not a known or established venue

There are options to “pay to win”

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Marketing-like soliciting of submissions

What about Journals

- Impact factor
- Prestigious ones in all fields: e.g. Nature, Science, .. Publishers with many journals with different impact factors.
- Computer Science : ACM, IEEE ..
- Longer review period (Desk reject - Under review -> Major corrections, Minor corrections (Iterative process))
- Sometimes it can take a couple of years to get the paper published

10 minutes break